

Power BI – DAX Information Functions

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Data Analysis Expressions, or DAX, are formulas or expressions that are used for calculations and data analysis. These expressions consist of a number of different functions, operators, and constants that are all evaluated together as a single formula to produce the desired results (value or values). In BI tools like Power BI, DAX formulas are very helpful because they enable data analysts to make the best use possible of the data sets they already have.

DAX Functions

A DAX function is a previously created formula that computes using values passed to it as arguments. Function arguments can include a column reference, numbers, text, constants, references to other formulas or functions, or logical values like TRUE or FALSE. They must be given to the function in a specific order. The values contained in an argument are subjected to a specific operation by every function. A DAX formula is capable of supporting multiple arguments.

Types of DAX Functions

- Date and Time Functions
- Time Intelligence Functions
- Information Functions
- Logical Functions
- Mathematical and Trigonometric Functions
- Statistical Functions
- Text Functions
- Parent-Child functions
- Other functions
- Table functions

The different DAX function types are described above. A DAX function is a previously created formula that computes using values passed to it as arguments. We now look at DAX Information Functions,

DAX Information Functions

When given a cell or row as an argument, DAX information functions examine it and inform you of the value's compatibility with the expected type. If the value you reference contains an error, for instance, the ISERROR function returns TRUE. Here are the DAX Information Functions with Output. In order to study these functions, you may look at the [dataset](#). Below is the screenshot of the dataset:

Order_ID				
A	B	C	D	
Order_ID	Customer_ID	Furniture	OnTime Delivery	Product
1001	9001	Sofas, Couches, Bookshelves, Tables	Yes	No
1002	9002	Sofas, Study Tables, Hammocks, Dining Tables	No	Yes
1003	9007	Bookshelves, Storage Furniture, Dining Tables	Yes	Yes
1004	9100	Dressing Tables, Folding Chairs, Shoe Racks, Study Tables	No	Yes
1005	9112	Dining Chairs, Folding Chairs, Wardrobes Sofas	Yes	No
1006	9220	sofas, Couches, Dining Chairs, Tables	Yes	No

Contains

The function either returns true or false depending on whether the values for all of the referred columns exist or are contained in those columns. If the values for all the referred columns are not contained, the function returns false.

Syntax: CONTAINS(<table>, <column value> <value>[<column value>, <value>]...)

Example :

The screenshot shows the Microsoft Power BI Desktop interface. The formula bar at the top displays the DAX formula: `contains_ex1 = CONTAINS('Fact', 'Fact'[OnTimeDelivery], "Yes", 'Fact'[ProductReturned], "yes")`. Below the formula bar, a table visualization displays the results of the query. The table has columns: OrderID, CustomerID, Furniture, OnTime Delivery, ProductReturned, and contains_ex1. The data rows correspond to the table shown in the first image, with the 'contains_ex1' column showing 'True' for rows 1003 and 1004, and 'False' for all other rows. The 'Total' row shows 'True'.

OrderID	CustomerID	Furniture	OnTime Delivery	ProductReturned	contains_ex1
1001	9001	Sofas, Couches, Bookshelves, Tables	Yes	No	False
1002	9002	Sofas, Study Tables, Hammocks, Dining Tables	No	Yes	False
1003	9007	Bookshelves, Storage Furniture, Dining Tables	Yes	Yes	True
1004	9100	Dressing Tables, Folding Chairs, Shoe Racks, Study Tables	No	Yes	True
1005	9112	Dining Chairs, Folding Chairs, Wardrobes Sofas	Yes	No	False
1006	9220	sofas, Couches, Dining Chairs, Tables	Yes	No	False
Total					True

CONTAINS – DAX Information Function

ContainsString

Depending on whether one string contains another, this function returns true or false.

Syntax: CONTAINSSTRING(<within text>, <find text>)

Example:

CONTAINSSTRING(SELECTEDVALUE('Fact'[Furniture]), 'Sofas')

OrderID	CustomerID	Furniture	OnTime Delivery	ProductReturned	contains_ex1	containsstring_ex2
1001	9001	Sofas, Couches, Bookshelves, Tables	Yes	No	False	True
1002	9002	Sofas, Study Tables, Hammocks, Dining Tables	No	Yes	False	True
1003	9007	Bookshelves, Storage Furniture, Dining Tables	Yes	Yes	True	False
1004	9100	Dressing Tables, Folding Chairs, Shoe Racks, Study Tables	No	Yes	False	False
1005	9112	Dining Chairs, Folding Chairs, Wardrobes, Sofas	Yes	No	False	True
1006	9220	sofas, Couches, Dining Chairs, Tables	Yes	No	False	True
Total					True	False

CONTAINSSTRING– DAX Information Function

ContainsStringExact

Whether one string contains another is indicated by a return value of TRUE or FALSE.

Syntax: CONTAINSSTRINGEXACT(<within_text>, <find_text>)

Example:

CONTAINSSTRINGEXACT(SELECTEDVALUE('Fact'[Furniture]), 'Sofas')

OrderID	CustomerID	Furniture	OnTime Delivery	ProductReturned	contains_ex1	containsstring_ex2	containsStringExact_ex3
1001	9001	Sofas, Couches, Bookshelves, Tables	Yes	No	False	True	True
1002	9002	Sofas, Study Tables, Hammocks, Dining Tables	No	Yes	False	True	True
1003	9007	Bookshelves, Storage Furniture, Dining Tables	Yes	Yes	True	False	False
1004	9100	Dressing Tables, Folding Chairs, Shoe Racks, Study Tables	No	Yes	False	False	False
1005	9112	Dining Chairs, Folding Chairs, Wardrobes, Sofas	Yes	No	False	True	True
1006	9220	sofas, Couches, Dining Chairs, Tables	Yes	No	False	True	False
Total					True	False	False

CONTAINSSTRINGEXACT – DAX Information Function

IsBlank

If the value is blank, this function returns true;

Syntax: ISBLANK(<Value>)

Example:

The screenshot shows the Power BI Desktop interface with the 'Table tools' and 'Column tools' tabs. The formula bar displays the DAX function `ISBLANK(Sheet1[Name])`. The table structure shows columns: Name, Id, and IS_BLANK_NAME. The data rows are:

Name	Id	IS_BLANK_NAME
Text1	1	False
Text2	2	False
	0	True

A red arrow points to the 'Output' column header, and a yellow box highlights the formula bar and the table data.

ISBLANK – DAX Information Function

IsNumber

This function determines whether a value is a number before returning either true or false.

Syntax: ISNUMBER(<value>)

Example:

The screenshot shows the Power BI Desktop interface with the 'Table tools' and 'Column tools' tabs. The formula bar displays the DAX function `ISNUMBER(Sheet1[Id])`. The table structure shows columns: Name, Id, IS_BLANK_NAME, IS_BLANK_ID, IF_BLANK_NAME, IF_BLANK_ID, and IS_NUMBER. The data rows are:

Name	Id	IS_BLANK_NAME	IS_BLANK_ID	IF_BLANK_NAME	IF_BLANK_ID	IS_NUMBER
Text1	1	False	False	False	False	True
Text2	2	False	False	False	False	True
	0	True	True	True	True	False

A red arrow points to the 'Output' column header, and a yellow box highlights the formula bar and the table data.

ISNUMBER – DAX Information Function

The DAX Information Functions with examples are shown above.

Here is the list of DAX Information Functions:

Functions	Description	Syntax
CONTAINS	The function either returns true or false depending on whether the values for all of the referred columns exist or are contained in those columns.	CONTAINS(<table>, <column value> <value>[, <column value>, <value>]...)
COLUMNSATISTICS	This function provides a statistics table for each column in each table in the model.	COLUMNSATISTICS()
CONTAINSROW	The function returns true if a row value exists or is contained in a table, or false otherwise.	CONTAINSROW(<tableExpression>, <ScalarExpression>[, <ScalarExpression>,...])
CONTAINSSTRING	Depending on whether one string contains another, this function returns true or false.	CONTAINSSTRING(<within text>, <find text>)
CONTAINSSTRINGEXACT	Whether one string contains another is indicated by a return value of TRUE or FALSE.	CONTAINSSTRINGEXACT(<within_text>, <find_text>)
CUSTOMDATA	returns the information found in the connection string's CustomData property.	CUSTOMDATA()
HASONEFILTER	If there are exactly One directly filtered values on ColumnName, the function either returns true or returns false.	HASONEFILTER(<column name>)
HASONEVALUE	If only one distinct value remains in the context for ColumnName, the function returns true; otherwise, it returns false.	HASONEVALUE(<columnName>)
ISBLANK	If the value is blank, this function returns true;	ISBLANK(<Value>)

Functions	Description	Syntax
ISERROR	If the value is an error, this function returns true; otherwise, it returns false.	ISERROR(<value>)
ISLOGICAL	This function determines if a value is logical, i.e. TRUE/FALSE: If true, it returns; if false, it returns	ISLOGICAL(<value>)
ISNUMBER	This function determines whether a value is a number before returning either true or false.	ISNUMBER(<value>)
ISNONTEXT	This function determines whether a value is nontext (blanks are treated as non-text), in which case it returns true or false.	ISNONTEXT(<value>)
ISTEXT	This function determines whether a value is text before returning true or false.	ISTEXT(<value>)
ISAFTER	a boolean function that behaves like a Start At clause and returns true for a row that satisfies all of the conditional criteria.	ISAFTER(<scalar_expression>, <scalar_expression>[, sort_order [, <scalar_expression>, <scalar_expression>[, sort_order]]...])
ISCROSSFILTERED	When columnName or another column in the same or related table is being filtered, this function returns TRUE.	ISCROSSFILTERED(<TableNameOrColumnName>)
ISEMPTY	verifies whether a table is empty.	ISEMPTY(<table_expression>)
ISEVEN	If the number is even, it returns TRUE; otherwise, it returns FALSE.	ISEVEN(number)

Functions	Description	Syntax
ISFILTERED	when columnName is being directly filtered, returns TRUE.	ISFILTERED(<TableNameOrColumnName>)
ISINSCOPE	When a column is the level in a hierarchy of levels, this function returns true.	ISINSCOPE(<columnName>)
ISODD	Returns FALSE if the number is even, or TRUE if the number is odd.	ISODD(<value>)
ISONORAFTER	a boolean function that behaves like a Start At clause and returns true for a row that satisfies all of the conditional criteria.	ISONORAFTER(<scalar_expression>, <scalar_expression>[, sort_order [, <scalar_expression>, <scalar_expression>[, sort_order]]...])
ISSELECTEDMEASURE	Expressions for calculation items use this information to determine whether the measure being referenced is one of the ones listed in the list of measures.	ISSELECTEDMEASURE(M1, M2, ...)
ISSUBTOTAL	Creates a new column in a SUMMARIZE expression that returns True if the row has values for the subtotal column specified as an argument and False otherwise.	ISSUBTOTAL(<columnName>)
NONVISUAL	identifies a value filter as non-visual in a SUMMARIZECOLUMNS expression.	NONVISUAL(<expression>)
SELECTEDMEASURE	Used in expressions for calculation items to refer to the currently referenced measure.	SELECTEDMEASURE()

Conclusion

This was a brief overview of Power BI's DAX and DAX Information Functions. Power BI is a very potent tool created by Microsoft that aids in the formulation of important business decisions and encourages the expansion of the company. By using these features, our information will be converted into a format that is simple to understand.

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